

Mod-2 periodicity for a family of sixth-order linear recurrences, with a matrix-functional generalization (Lean 4 machine-checked claim of record)

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Abstract

We record two theorems about the sixth-order linear recurrence $c(n+6) = 4c(n+5) + 14c(n+4) - 10c(n+2) + c(n)$. First, every integer sequence satisfying it has parity of period dividing 6, and the parity of $c(n)$ is determined by $n \bmod 6$. Second, a matrix-functional generalization: any 8×8 integer matrix T annihilated by $x^6 - 4x^5 - 14x^4 + 10x^2 - 1$ yields scalar sequences $n \mapsto (T^n v)_i$ obeying the same recurrence. The motivation comes from monomer–dimer (matching) enumeration on $3 \times n$ boards with one boundary defect, where the full-board counts are OEIS A033506 and the defect-family sequences behind this recurrence are recorded here for the first time; that combinatorial interpretation is *not* part of the machine-checked claim. All proofs are machine-checked by the Lean 4 kernel (0 `sorry`; axioms within `{propext, Quot.sound, Classical.choice}`). A structured novelty search found no occupying record. The artifact package is deposited on Zenodo. This note is a priority claim of record, not a full paper.

1 Introduction

Counting monomer–dimer coverings of a $3 \times n$ board — matchings of the grid graph $P_3 \times P_n$ — is a classical enumeration problem: the full-board counts form OEIS sequence A033506 [1], with a generating function documented there following Lundow’s enumeration of matchings in polygraphs [2, 3]; see Propp’s survey for the landscape [4]. Forcing one prescribed boundary cell to carry a monomer (deleting one vertex, in matching language) produces

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defect counting sequences. The pipeline’s transfer-matrix computations of three such families — a side-middle hole, a top-edge hole, and a fixed edge cut — produce integer sequences that, as of September 2026, do not appear in the OEIS under calibrated exact-term searches, and whose recurrences share the sixth-order signature $(4, 14, 0, -10, 0, 1)$ of the full board. The nearest literature studies *perfect* matchings of vertex-deleted lattices (pure dimer coverings), on odd-by-odd boards [5, 6, 7]; adjacent non-occupying works prove a reciprocity theorem for total matchings of full boards [8] and study the matching complex of the same $3 \times n$ family topologically [9].

This note records two theorems at the recurrence level, produced by AI4Math, a highly automated LLM-plus-kernel pipeline in which the Lean 4 kernel is the sole arbiter of correctness. The present text is a priority claim of record — a short deposit note accompanying a Zenodo artifact package; a fuller narrative paper is prepared separately through the pipeline’s paper lane. Every theorem stated below is a *complete proof* in the pipeline’s grading: it compiles with 0 **sorry**, its axioms stay within the standard whitelist, and the statement was frozen before proving. The combinatorial reading of the recurrence (tiling counts) is motivation only; Section 6 is explicit about what the kernel did *not* prove.

2 Statement

The formal development adds no definitions: each theorem is self-contained, quantifying over the objects in its hypothesis. The frozen statement files contain exactly one theorem each (`edgemid_master` in `edgemid-p1-statement.lean`, sha256 `f964ce71...27b27c56`; `edgemid_matrix_family` in `edgemid-p2-statement.lean`, sha256 `cf9e379a...99b6d5`). The formal theorem headers, including their Chinese docstrings as frozen, appear verbatim inside Listing 1 (lines 15–20 and 44–52 there), byte-identical to the frozen files.

Theorem 1 (mod-2 periodicity and characterization). *Let $c : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy $c(n+6) = 4c(n+5) + 14c(n+4) - 10c(n+2) + c(n)$ for all $n \in \mathbb{N}$. Then:*

1. $c(n+6) \equiv c(n) \pmod{2}$ for all n ; and
2. $c(n) \equiv c(n \bmod 6) \pmod{2}$ for all n , i.e. the parity of $c(n)$ is completely determined by $n \bmod 6$.

Theorem 2 (matrix-functional generalization). *Let T be an 8×8 integer matrix satisfying $T^6 = 4T^5 + 14T^4 - 10T^2 + I$, let $v \in \mathbb{Z}^8$ and let i be an*

index. Then for all $n \in \mathbb{N}$,

$$(T^{n+6}v)_i = 4(T^{n+5}v)_i + 14(T^{n+4}v)_i - 10(T^{n+2}v)_i + (T^n v)_i,$$

i.e. the scalar sequence $n \mapsto (T^n v)_i$ satisfies the recurrence of Theorem 1.

3 Proof

The complete machine proofs are reproduced verbatim in Listing 1, which is the whole frozen final development `final/03-proof-complete.lean` (66 lines, sha256 1124e6a5...72c1d): imports, header comment, both theorems with their full proofs. In brief, Theorem 1 converts the congruence into a divisibility via `Int.modEq_iff_dvd`, discharges the parity rearrangement with the recurrence hypothesis and `ring`, lifts the one-step congruence to six-step blocks by induction (`Int.ModEq.refl`, `Int.ModEq.trans`), and finishes with the division identity $n = n \bmod 6 + 6\lfloor n/6 \rfloor$ (`omega`). Theorem 2 rewrites $T^{n+6} = T^n T^6$ (`pow_add`), substitutes the annihilating relation, distributes over the matrix operations (`Matrix.add_mulVec`, `Matrix.sub_mulVec`, `Matrix.smul_mulVec`, `mul_smul_comm`), and closes by `rfl` along a finite fallback chain. A narrative prose proof is not the duty of this note; it belongs to the full paper lane.

Listing 1: Complete frozen final development (verbatim full text, lines 1–66 of `final/03-proof-complete.lean`, sha256 1124e6a5...72c1d). The two theorem headers (lines 15–20, 44–52) are byte-identical to the frozen statement files; docstrings and the header comment are in Chinese as frozen.

```

1  import Mathlib.Data.Int.ModEq
2  import Mathlib.Data.Matrix.Mul
3
4  /-!
5  # 03 证明完成稿（候选 03 · edgemid-pipeline · P1+P2）
6
7  两定理 statement 头与冻结快照逐字一致（宪条 2）：P1 锚 `formalized/
   edgemid-p1-statement.lean:12-17`,
8  P2 锚 `formalized/edgemid-p2-statement.lean:14-22`（两者均锚定
9  `tasks/20260926-beian-extselect/candidates/smoke-03/
   EdgemidSmoke03r3.lean` 定理头）。
10 proof body 逐字取自 smoke r3（P1 :18-39 / P2 :50-63, 逐字节一致零改
   动），
11 全稿零占位符、无 namespace。imports = 两冻结件 import 面并集。
12 AI 生成（2026-09-26 军政部工人，dynamic-workflow 子代理）。
13 -/

```

```

14
15 /-- P1(首选命题):凡满足六阶线性递推
16 `c (n+6) = 4 * c (n+5) + 14 * c (n+4) - 10 * c (n+2) + c n` 的整数
    序列,
17 模 2 以 6 为周期,且模 2 余数由 `n % 6` 完全决定(整除刻画支)。 -/
18 theorem edgemid_master (c : ℕ → ℤ)
19   (h : ∀ n, c (n+6) = 4 * c (n+5) + 14 * c (n+4) - 10 * c (n+2)
    + c n) :
20   (∀ n, c (n+6) ≡ c n [ZMOD 2]) ∧ (∀ n, c n ≡ c (n % 6) [ZMOD
    2]) := by
21   have key : ∀ n, c (n+6) ≡ c n [ZMOD 2] := by
22     intro n
23     refine Int.modEq_iff_dvd.2 ⟨-(2 * c (n+5) + 7 * c (n+4) - 5 *
    c (n+2)), ?_⟩
24     rw [h n]
25     ring
26   have aux : ∀ k n, c (n + 6 * k) ≡ c n [ZMOD 2] := by
27     intro k
28     induction k with
29     | zero =>
30       intro n
31       simp only [Nat.mul_zero, Nat.add_zero]
32       exact Int.ModEq.refl _
33     | succ k ih =>
34       intro n
35       have e : n + 6 * (k+1) = (n + 6 * k) + 6 := by ring
36       rw [e]
37       exact (key (n + 6 * k)).trans (ih n)
38   refine ⟨key, fun n => ?_⟩
39   have step : c (n % 6 + 6 * (n / 6)) ≡ c (n % 6) [ZMOD 2] := aux
    (n / 6) (n % 6)
40   have e : n % 6 + 6 * (n / 6) = n := by omega
41   rw [e] at step
42   exact step
43
44 /-- P2(挑战档扩展):转移矩阵被六阶多项式零化时,其矩阵幂泛函
45 (含各固定孔位缺陷计数)满足同一递推。 -/
46 theorem edgemid_matrix_family {T : Matrix (Fin 8) (Fin 8) ℤ}
47   (v : Fin 8 → ℤ) (i : Fin 8)
48   (hT : T ^ 6 = 4 • T ^ 5 + 14 • T ^ 4 - 10 • T ^ 2 + (1 :
    Matrix (Fin 8) (Fin 8) ℤ))
49   (n : ℕ) :
50   (T ^ (n+6)).mulVec v i
51   = 4 * (T ^ (n+5)).mulVec v i + 14 * (T ^ (n+4)).mulVec v i
52   - 10 * (T ^ (n+2)).mulVec v i + (T ^ n).mulVec v i := by

```

```

53 have hgen : T ^ (n+6)
54   = 4 • T ^ (n+5) + 14 • T ^ (n+4) - 10 • T ^ (n+2) + (T^n :
    Matrix (Fin 8) (Fin 8) ℤ) := by
55   rw [pow_add, hT, Matrix.mul_add, Matrix.mul_sub, Matrix.
    mul_add,
56     mul_smul_comm, mul_smul_comm, mul_smul_comm,
57     Matrix.mul_one, ← pow_add, ← pow_add, ← pow_add]
58   rw [hgen, Matrix.add_mulVec, Matrix.sub_mulVec, Matrix.
    add_mulVec,
59     Matrix.smul_mulVec, Matrix.smul_mulVec, Matrix.smul_mulVec]
60   first
61   | rfl
62   | simp only [smul_eq_mul]
63   | simp [smul_eq_mul]
64   | norm_num
65   | push_cast <;> rfl
66   | lia

```

4 Verification evidence

- Toolchain: Lean v4.34.0 (leanprover/lean4:v4.34.0) [10], mathlib4 v4.34.0 (rev 5ed29652) [11].
- Statement integrity: frozen statements formalized/edgemid-p1-statement.lean, sha256 f964ce71e9888d7e8d9d982996fed2531492afdb1ca47df004a7f5d727b27c56, and formalized/edgemid-p2-statement.lean, sha256 cf9e379af46affeee0306ba2bb84094a18, final development final/03-proof-complete.lean, sha256 1124e6a53b6ec70fee5c964490fa4f, symmetric difference of the two theorem headers against the frozen snapshots (frozen lines 12–17 vs final lines 15–20; frozen lines 14–22 vs final lines 44–52): zero, byte-identical.
- Kernel check: strict compile exit 0 with no errors; a byte-level scan of the whole final file (comments included) finds 0 occurrences of sorry/admit. Gate-2 non-degeneracy: all seven bare-tactic probes (nlinarith, norm_num, simp, ring_nf, positivity, omega, linarith) fail to close either goal. Axiom audit, verbatim from audit/final-axioms-p1.txt and audit/final-axioms-p2.txt:

```

WELLDEF OK
AXIOMS edgemid_master [propext, Quot.sound]

WELLDEF OK

```

AXIOMS edgemid_matrix_family [propext, Classical.choice, Quot.sound]

Both lists are subsets of the whitelist {propext, Quot.sound, Classical.choice}; `sorryAx` is absent.

- Audit chain: statement freeze with sha256 snapshots; gate-2 well-definedness compilation plus degeneracy probing; proof assembly restricted to proof bodies; gate-3 final strict compilation, byte-level symmetric-difference comparison and axiom audit; an NL–Lean roundtrip faithfulness walkthrough at formalization time. Evidence originals are in the artifact package `audit/`.

5 Novelty evidence

The pipeline’s C9 novelty review adjudicated this result as *no occupying record found*, and an independent exclusivity re-review (2026-09-26, before deposit) maintained that adjudication. This is a statement about search evidence, not a claim to new mathematics: the full-board object A033506 is an occupied OEIS record [1], and the three defect-family sequences are recorded here for the first time at the level of calibrated exact-term search evidence (wording capped accordingly). Query channels and query-string summary (the full query log, including zero-hit evidence files, is in `audit/c9-record.md` of the package, copied verbatim from the source record):

channel	query summary	result
OEIS sequence layer	calibration: baseline 8 terms 1, 3, 22, 131, 823, 5096, 31687, 196785	hits A033506 (channel works)
OEIS sequence layer	exact 8-term W / N1 / EB0; relaxed mid-window 5-term variants; corner- hole control 8 terms	all null (zero hits)
OEIS record layer	six full-text queries (monomer-dimer; deleted vertex + matchings; grid graph + matchings; square removed + 3 x n; monomer + 3 x n; matchings + one vertex); A033506 cross-references	hits are whole- board families or tatami en- tries; no $3 \times n$ defect variant
literature layer	arXiv (7 query groups, 8 abstracts verified, 1 full text checked), Ope- nAlex (6 groups), zbMATH (5 groups), Semantic Scholar (1 success, 2 rate-limited)	no occupying record; nearest line [7, 5, 6] demarcated, not occupying
web layer	general web, English and Chinese sides	whole-board classics and OI/program- ming tutorials only
library layer	mathlib ripgrep for monomer / dimer	zero hits

The nearest literature line — Oh’s state matrix recursion method [7] and the Tzeng–Wu / Wu Pfaffian results [5, 6] — counts *perfect* matchings of vertex-deleted lattices (pure dimer coverings), the latter two restricted to odd-by-odd boards; the sequences behind this note count *all* matchings of the vertex-deleted $3 \times n$ graph, for all n . Different object, no occupation; cited here for demarcation. Recorded residual gaps, verbatim in spirit from the re-review: (i) CNKI (Chinese-language database) was not searched (login required); (ii) two Semantic Scholar queries were rate-limited (HTTP 429), with one query succeeding; (iii) the Tzeng–Wu 2003 and Wu 2006 original PDFs were not fetched — the demarcation rests on the ar5iv full text of [7] plus abstracts.

6 Scope and limitations

Grading boundary, per the final audit: the two theorems are complete proofs, strictly as their Lean statements — Theorem 1 about *arbitrary* inte-

ger sequences satisfying the recurrence, and Theorem 2 as a uniform recurrence for matrix functionals of *arbitrary* annihilated 8×8 integer matrices. Not machine-checked here: (i) any identification of a concrete T with the transfer matrix of a $3 \times n$ defect board, and any equality of $(T^n v)_i$ with a defect matching count — that bridge is the standard transfer-matrix narrative [1, 3, 4], and mathlib currently ships no monomer–dimer or matching-enumeration API to build it on (an eight-item gap list, ripgrep-verified, is in the artifact’s evidence pack); (ii) the claim that the W / N1 / EB0 defect-family sequences satisfy the recurrence at all, which stands at the level of numerical evidence (term-window fits) plus, for the full board, the OEIS-documented generating function — OEIS/literature endorsement and a computational conjecture layer, not kernel content: the kernel proved no counting ontology. The periodicity branch of Theorem 1 is algebraically shallow (a one-line parity rearrangement); the characterization branch carries the content — stated to prevent over-reading. Novelty claims are search evidence on a specific date, not proofs of absence. This note contains no literature survey; it is a deposit record, not a survey article.

Data availability

Artifact package: Zenodo deposit (DOI to be reserved at upload; see UPLOAD.md in the source repository) with a public mirror on GitHub at <https://github.com/Aurora0134/ai4math-claims>. Note under CC BY 4.0, code under MIT.

Use of generative AI

(a) Stage mapping: candidate-proposition generation, formalization into Lean statements, retrieval evidence packing, proof assembly, audit documentation, and text drafting (including this note) were performed by AI (LLM) stages of the AI4Math pipeline; topic adjudication, statement-semantics verification and gate sign-off were human. (b) Model/node/budget (numbers copied verbatim from the source task ledger `tasks/20260926-edgemid-pipeline/budget.log`): a single user-designated model node, wire-id `anthropic/S3AI-Grok/grok-4.7` (model-detect resolution level L2; session runtime GLM-5.3 Flash via official direct connection, per the ledger’s node-snapshot line), no cross-node switching. Source task totals: llm-call 0/6 (cap 6, expected 0 achieved — all drafting ran on the session runtime model through workflow subagents, so the call-metered budget records zero and there are no pass-at-k rates to

report); Lean compilations 9/24 on the kernel-script caliber (18/24 including worker-side re-verification runs; both calibers are recorded line by line in the ledger); diagnostics 0/4; timebox about 2 hours 20 minutes within 24 hours. This claim-note stage used llm-call 0 of its budget 4, and 10 LaTeX compilations of its budget 6 (counting the final rebuild that embedded this number; ledger `claims/edgemid-monomer-dimer/budget.log`). (c) Final arbiter: all statements and proofs reported here were checked by the Lean 4 kernel: every proof compiles with no `sorry`, and `#print axioms` reports only subsets of `{propext, Quot.sound, Classical.choice}`. (d) The human author reviewed and is responsible for all content; the AI system is not an author.

Author contributions

CRediT-style: the human author was responsible for topic selection and adjudication, verification of statement semantics, gate sign-off, and the decision to deposit; the AI4Math pipeline (an LLM system) performed proposition generation, proof search, and text drafting. The AI system is not listed as an author.

References

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